CCNA Lab Series

Packet Tracer Edition — Junior Level

Author	Michael C
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Platform	Cisco Packet Tracer 8.x
Equipment	3× Cisco 2911 Routers • 3× 2960-24TT Switches • 3× PCs
Focus Areas	Static Routes • Default Routes • Floating Static Routes • Route Verification

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01 Lab Objective & Scope

Static routing is the foundational method by which a network administrator manually instructs a router where to forward traffic destined for networks it does not have directly connected interfaces on. While dynamic routing protocols (OSPF, EIGRP) automate this process at scale, understanding static routing is essential for troubleshooting, stub networks, and building a strong networking foundation.

In this lab, a three-router linear topology was built in Cisco Packet Tracer. Static routes were configured to establish full end-to-end connectivity between all three LAN segments. A default route was then implemented to simplify R1's routing table, and finally a floating static route was added to demonstrate path redundancy.

Skill Area	What Was Demonstrated
Static Routes	Manually defining next-hop paths to each remote network
Default Routes	Implementing a gateway of last resort (0.0.0.0/0)
Floating Static Routes	Configuring a backup path with elevated Administrative Distance
Verification	Using show ip route, ping, and tracert to validate connectivity

02 Network Topology

The lab topology consists of three Cisco 2911 routers connected in a linear chain, with each router serving a LAN segment via a 2960-24TT switch. WAN links between routers use /30 point-to-point subnets to conserve address space.

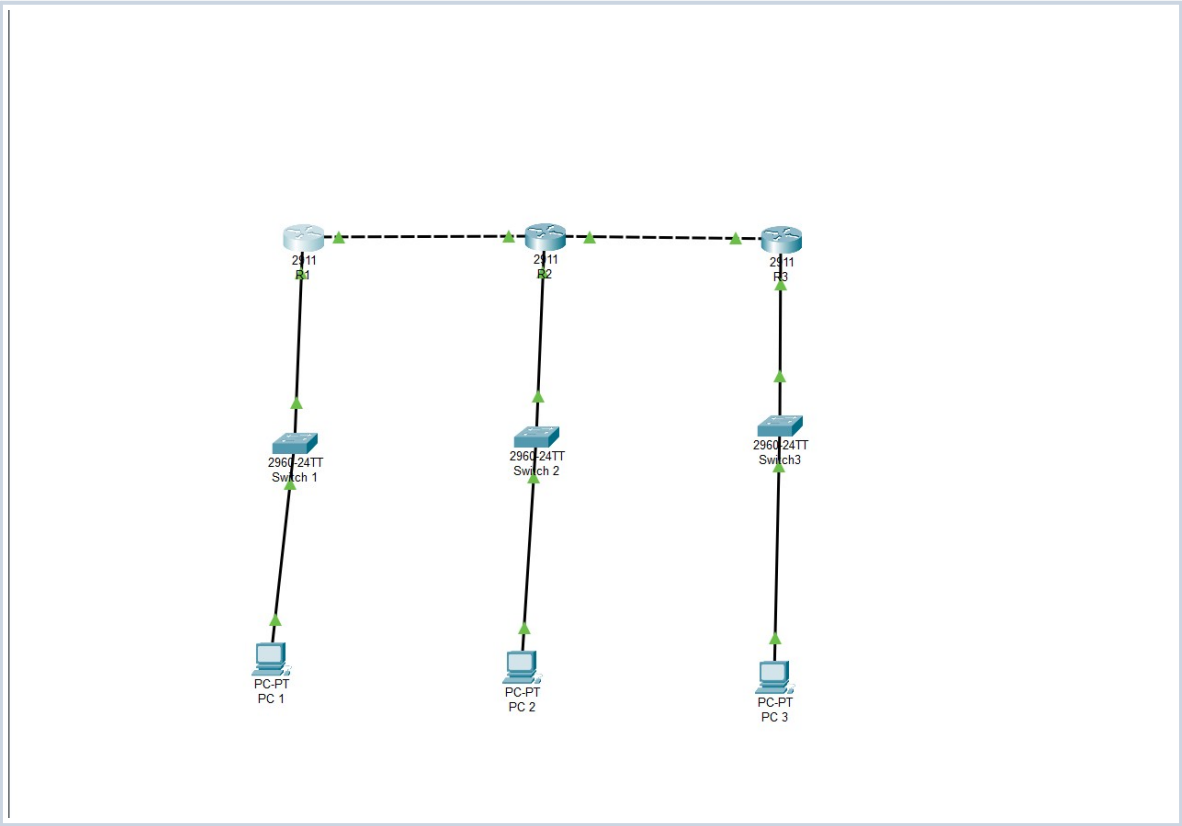


Figure 1 — Initial Packet Tracer topology: R1–R2–R3 linear chain with three LAN segments

All router-to-router links use GigabitEthernet interfaces with /30 subnets (only 2 usable host addresses per link). Each LAN uses a /24 subnet providing up to 254 hosts per segment.

03 IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Network / Role
R1	G0/0	192.168.12.1	/30	WAN link to R2
R1	G0/1	192.168.1.1	/24	LAN-A (PC1 gateway)
R2	G0/0	192.168.12.2	/30	WAN link to R1
R2	G0/1	192.168.23.1	/30	WAN link to R3
R2	G0/2	192.168.2.1	/24	LAN-B (PC2 gateway)
R3	G0/0	192.168.23.2	/30	WAN link to R2
R3	G0/1	192.168.3.1	/24	LAN-C (PC3 gateway)
PC1	NIC	192.168.1.10	/24	GW: 192.168.1.1
PC2	NIC	192.168.2.10	/24	GW: 192.168.2.1
PC3	NIC	192.168.3.10	/24	GW: 192.168.3.1

Subnet Breakdown: The /30 WAN subnets provide exactly 2 usable addresses each — ideal for point-to-point router links. The /24 LAN subnets each support up to 254 end hosts.

04 Step 1 — Router Interface Configuration

Before configuring any routes, all router interfaces must be assigned their IP addresses and brought online. Each router was accessed via the CLI tab and configured in global configuration mode.

R1 — Interface Configuration

R1 serves as the gateway for LAN-A (192.168.1.0/24) and connects to R2 via the 192.168.12.0/30 WAN link. Both G0/0 (WAN) and G0/1 (LAN) were configured with IP addresses and enabled.

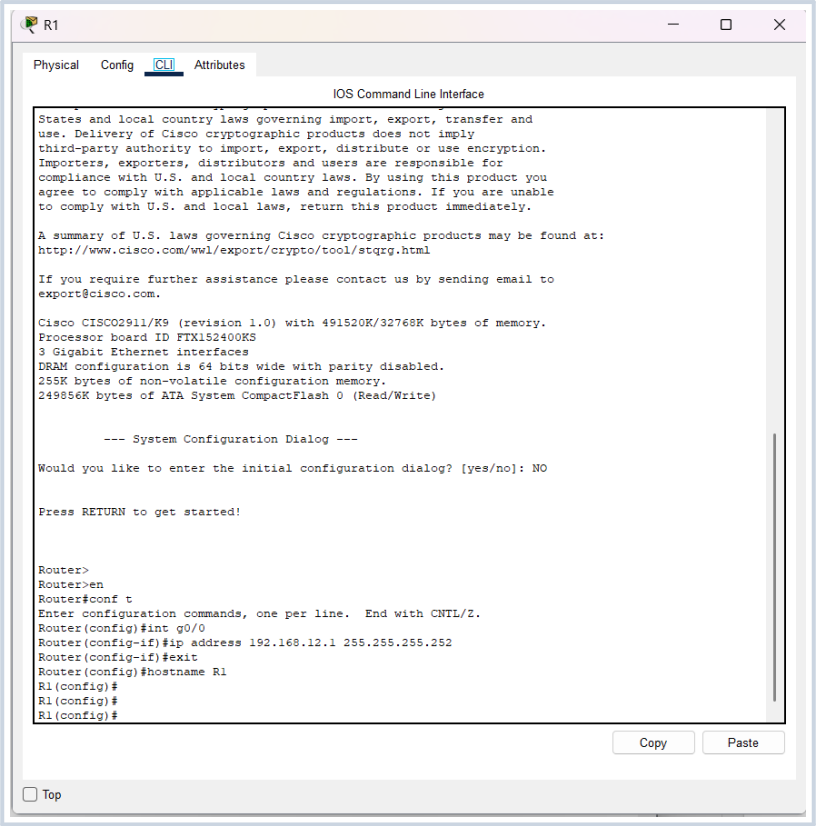


Figure 2 — R1 initial boot: hostname set, G0/0 configured with 192.168.12.1/30

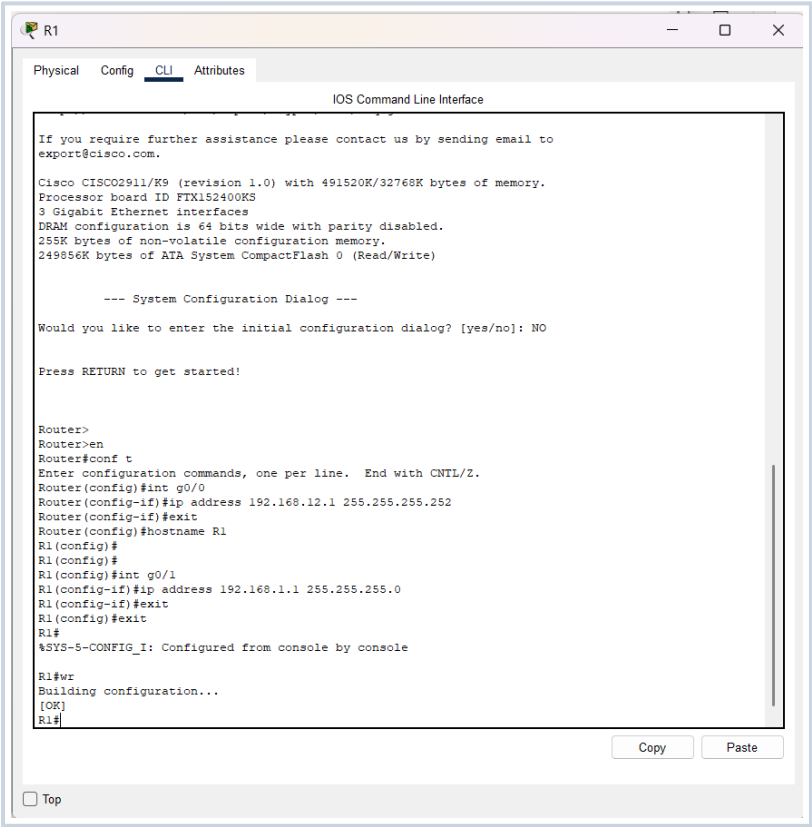


Figure 3 — R1 complete: G0/1 configured with 192.168.1.1/24, configuration saved with 'wr'

R2 — Interface Configuration

R2 is the central hub router with three active interfaces — two WAN links (to R1 and R3) and one LAN interface for LAN-B. Note the typo correction during configuration: an incomplete subnet mask was caught and re-entered correctly.

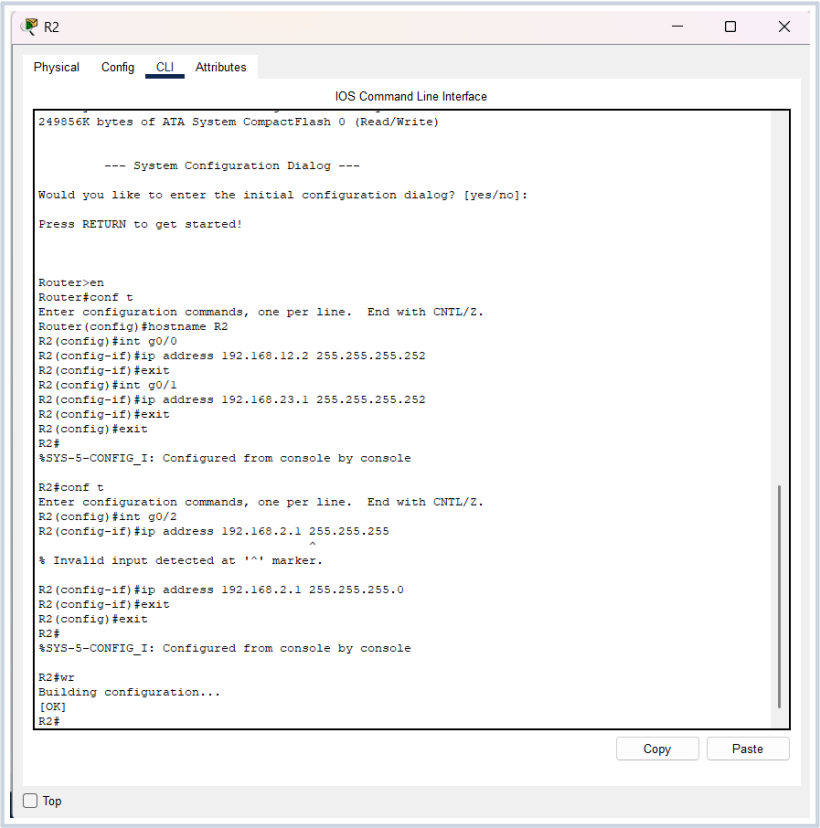


Figure 4 — R2 configuration: G0/0 (192.168.12.2/30), G0/1 (192.168.23.1/30), G0/2 (192.168.2.1/24)

Note: An invalid subnet mask was initially entered for G0/2 (`255.255.255` without the final octet). The IOS parser caught this error with `% Invalid input detected at '^' marker`. This was corrected to 255.255.255.0 — a good example of real-world troubleshooting.

R3 — Interface Configuration

R3 connects to R2 via the 192.168.23.0/30 WAN link and serves LAN-C (192.168.3.0/24).

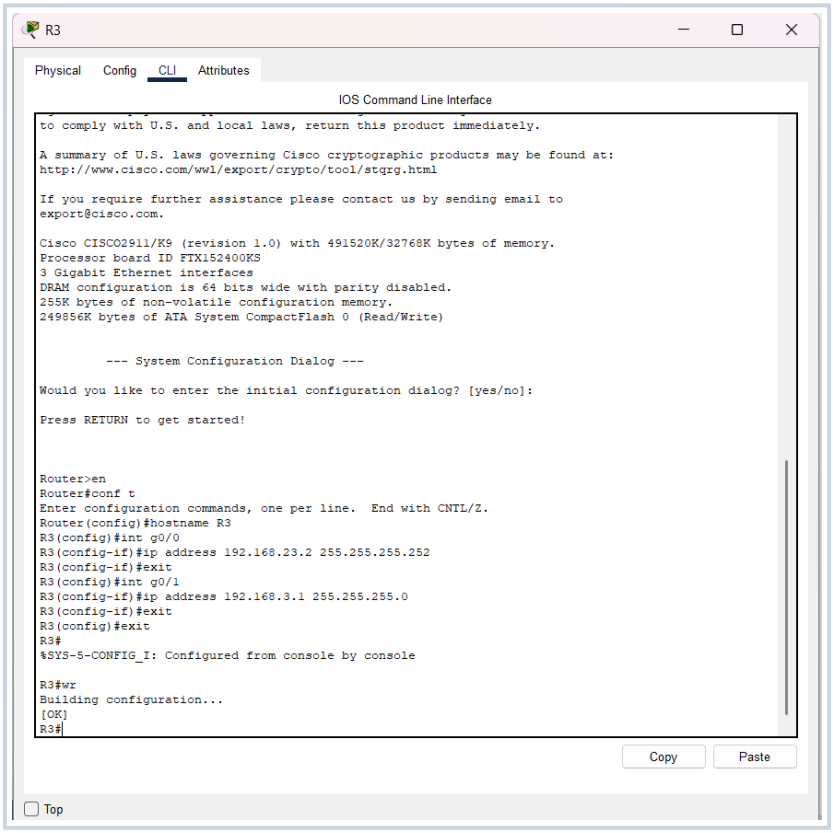


Figure 5 — R3 configuration: G0/0 (192.168.23.2/30), G0/1 (192.168.3.1/24), saved

05 Step 2 — Static Route Configuration

With all interfaces configured and active, each router only knows about its directly connected networks. To enable end-to-end communication, static routes must be added on each router pointing to every remote network it needs to reach.

R1 — Static Routes to Remote Networks

R1 needs routes to three remote networks: LAN-B (192.168.2.0/24), LAN-C (192.168.3.0/24), and the R2–R3 WAN link (192.168.23.0/30). All traffic is forwarded to R2 (next-hop 192.168.12.2).

```
R1(config)#ip route 192.168.2.0 255.255.255.0 192.168.12.2
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.12.2
R1(config)#ip route 192.168.23.0 255.255.255.0 192.168.12.2
```

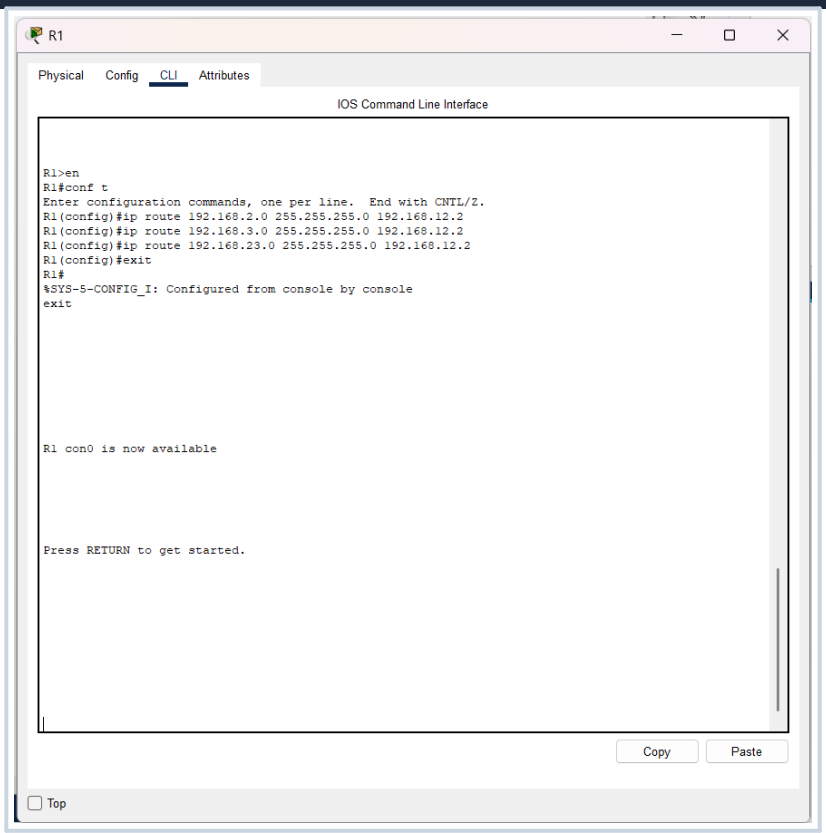


Figure 6 — R1: Three static routes configured, all pointing to next-hop 192.168.12.2 (R2)

R2 — Bidirectional Static Routes

As the middle router, R2 needs routes in both directions: toward LAN-A via R1 (192.168.12.1) and toward LAN-C via R3 (192.168.23.2).

```
R2(config)#ip route 192.168.1.0 255.255.255.0 192.168.12.1
R2(config)#ip route 192.168.3.0 255.255.255.0 192.168.23.2
```

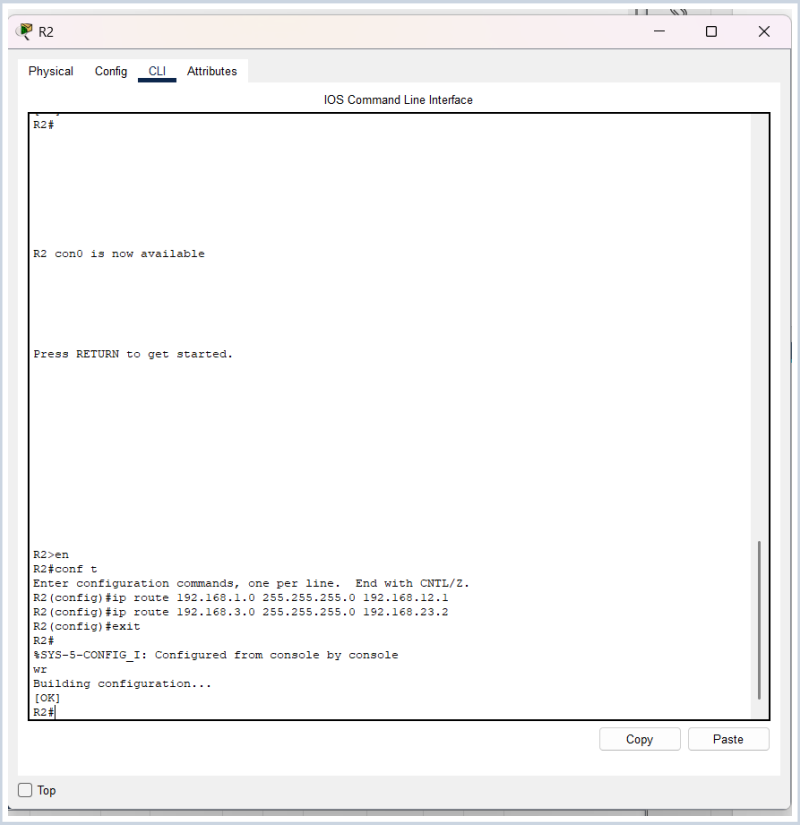


Figure 7 — R2: Two static routes configured for bidirectional reachability

R3 — Return Path Static Routes

R3 needs return routes to LAN-A, LAN-B, and the R1–R2 WAN link. All traffic is forwarded to R2 (next-hop 192.168.23.1).

```
R3(config)#ip route 192.168.1.0 255.255.255.0 192.168.23.1
R3(config)#ip route 192.168.2.0 255.255.255.0 192.168.23.1
R3(config)#ip route 192.168.12.0 255.255.255.252 192.168.23.1
```

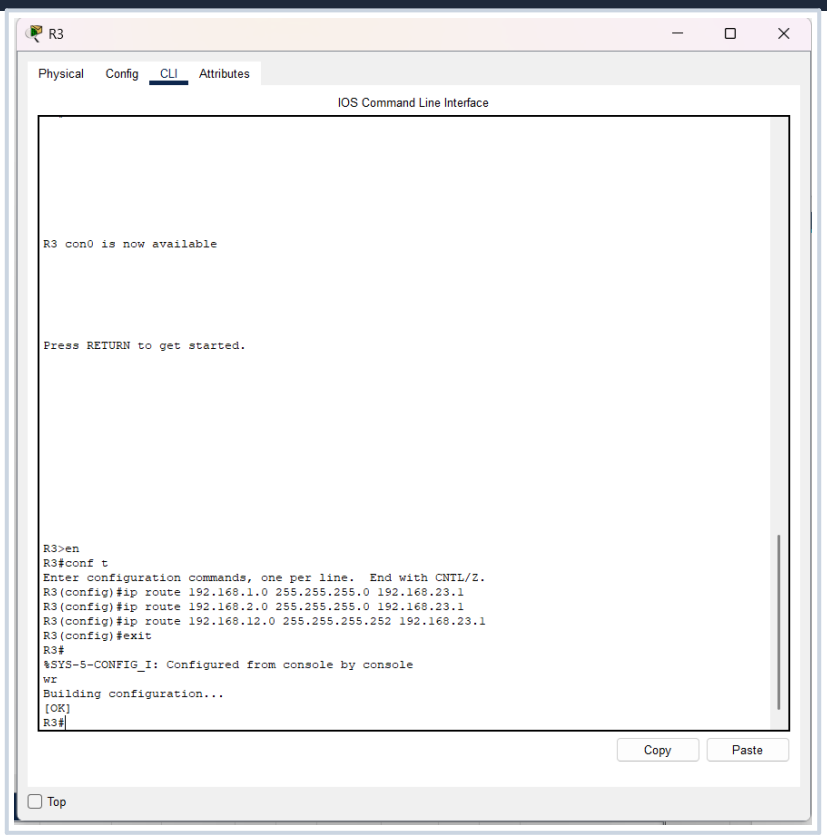


Figure 8 — R3: Three static routes configured, all pointing to next-hop 192.168.23.1 (R2)

Key Principle: Routing must be bidirectional. If R1 has a route to reach LAN-C but R3 lacks a return route to LAN-A, reply packets will be dropped. Every static route needs a corresponding return path.

06 Step 3 — Verification & Connectivity Testing

With all static routes in place, end-to-end connectivity was tested using ping and tracert from the PC command prompts. The routing table was also inspected on R1 to confirm all entries.

R1 — Routing Table Verification

The `show ip route` command on R1 displays both directly connected (C) and static (S) routes. Three static entries are visible, confirming the configuration was applied correctly.

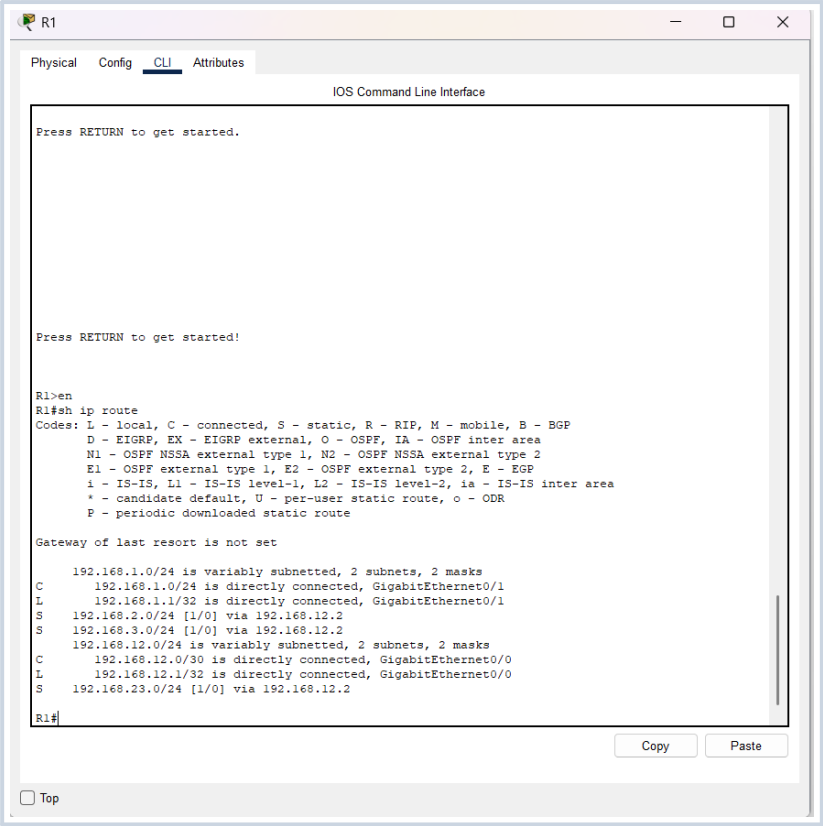


Figure 9 — R1 routing table: Connected (C/L) and Static (S) routes to all remote networks

PC1 — Ping Connectivity Tests

Initial ping attempts from PC1 to PC2 (192.168.2.2) showed 100% packet loss as ARP tables were being populated. Subsequent pings succeeded, confirming the static routes are functioning.

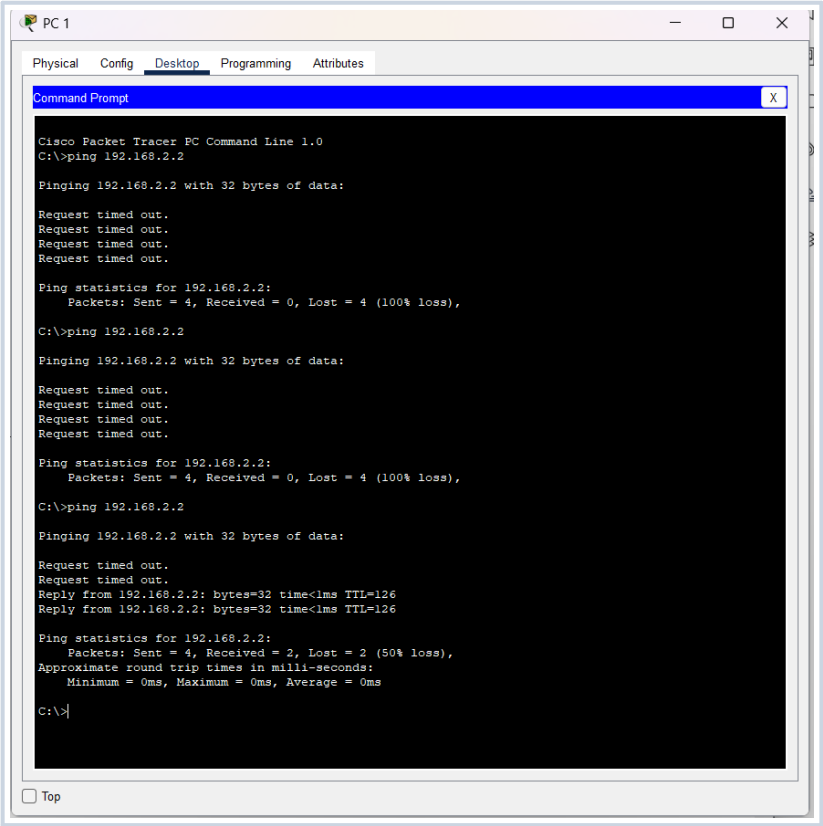


Figure 10 — PC1 initial pings: first attempts time out while ARP resolves, then replies arrive

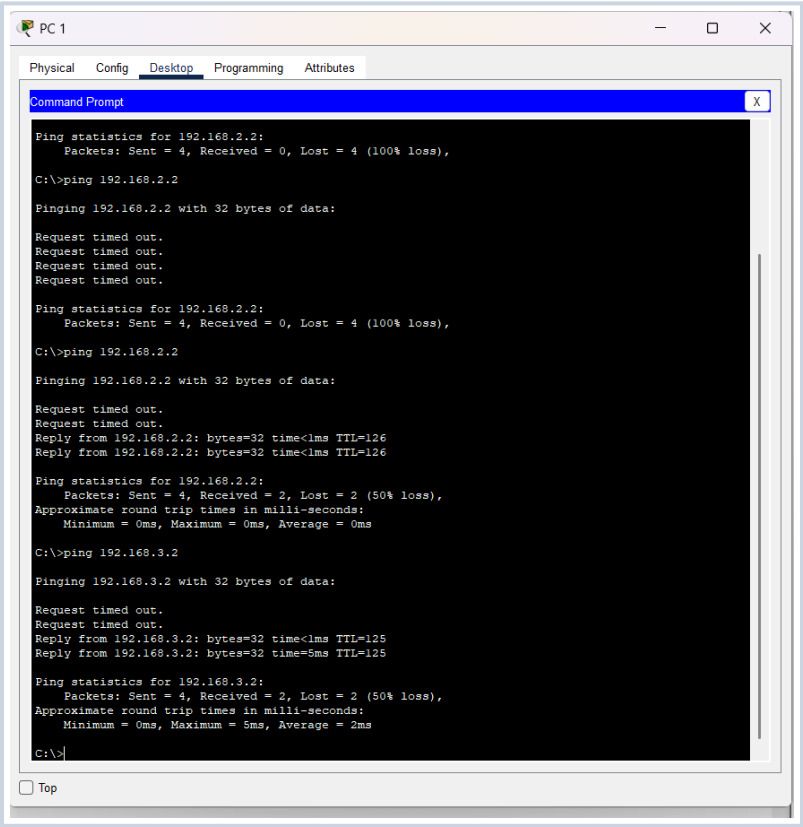


Figure 11 — PC1 successful pings to both PC2 (192.168.2.2) and PC3 (192.168.3.2)

Result: Connectivity confirmed — PC1 can reach both PC2 and PC3 across multiple router hops. The initial timeouts are normal ARP behavior, not a routing issue.

PC2 — Cross-Verification

To confirm bidirectional reachability, ping tests were also run from PC2 to PC1 (192.168.1.2) and PC3 (192.168.3.2). Both returned 0% loss — full mesh connectivity is operational.

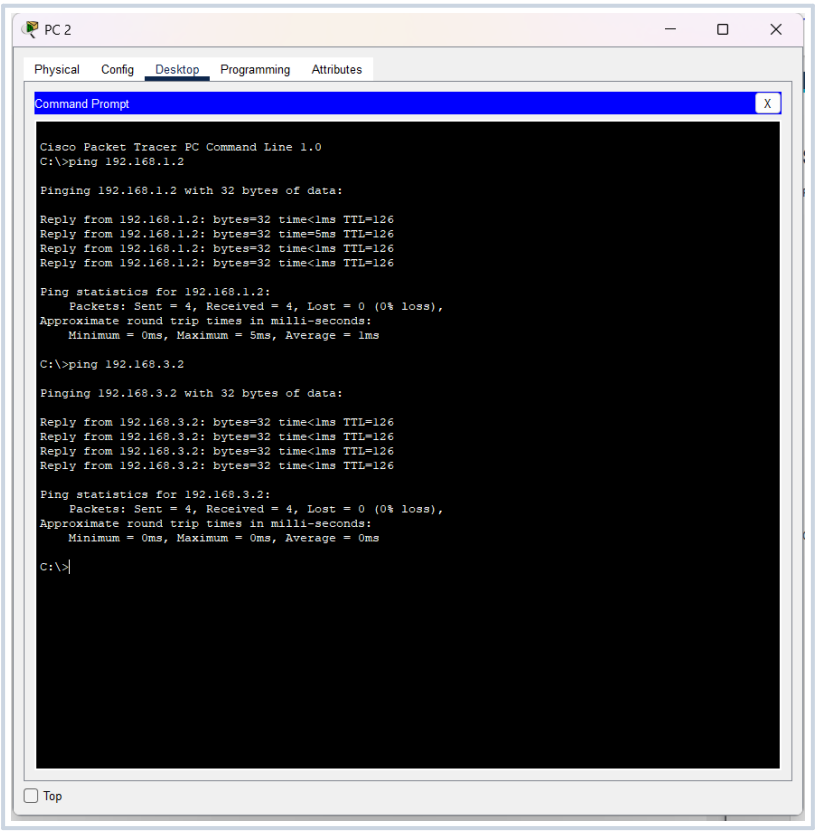


Figure 12 — PC2 pings to PC1 (192.168.1.2) and PC3 (192.168.3.2): 0% loss on both

Traceroute — Path Verification

A traceroute from PC1 to PC3 (192.168.3.2) reveals the exact forwarding path through the network. Four hops are shown: PC1's gateway (R1), the R1–R2 link, the R2–R3 link, and finally PC3.

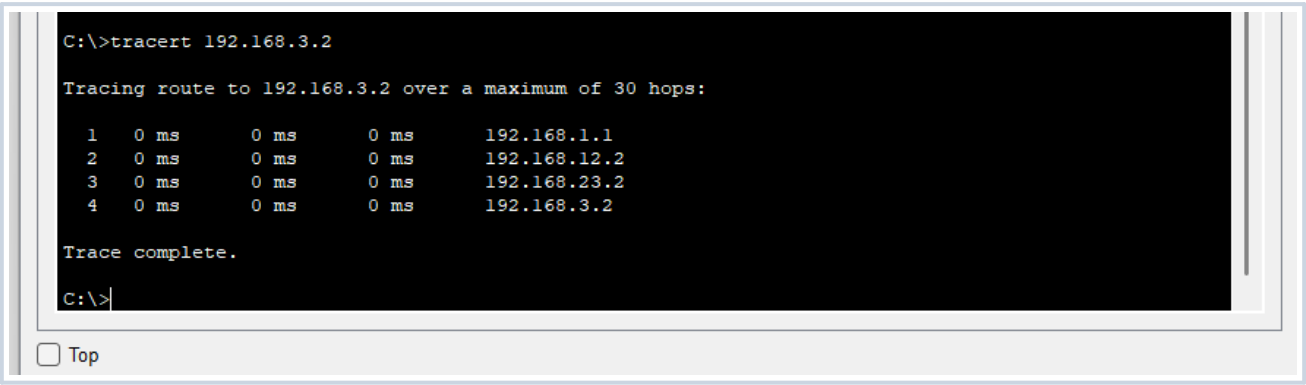


Figure 13 — Traceroute PC1 → PC3: 4 hops confirming path R1 (192.168.1.1) → R2 (192.168.12.2) → R3 (192.168.23.2) → PC3 (192.168.3.2)

Path Analysis: Hop 1 = R1 LAN gateway (192.168.1.1), Hop 2 = R2 WAN (192.168.12.2), Hop 3 = R3 WAN (192.168.23.2), Hop 4 = PC3 (192.168.3.2). All hops respond in <1ms, indicating healthy forwarding performance.

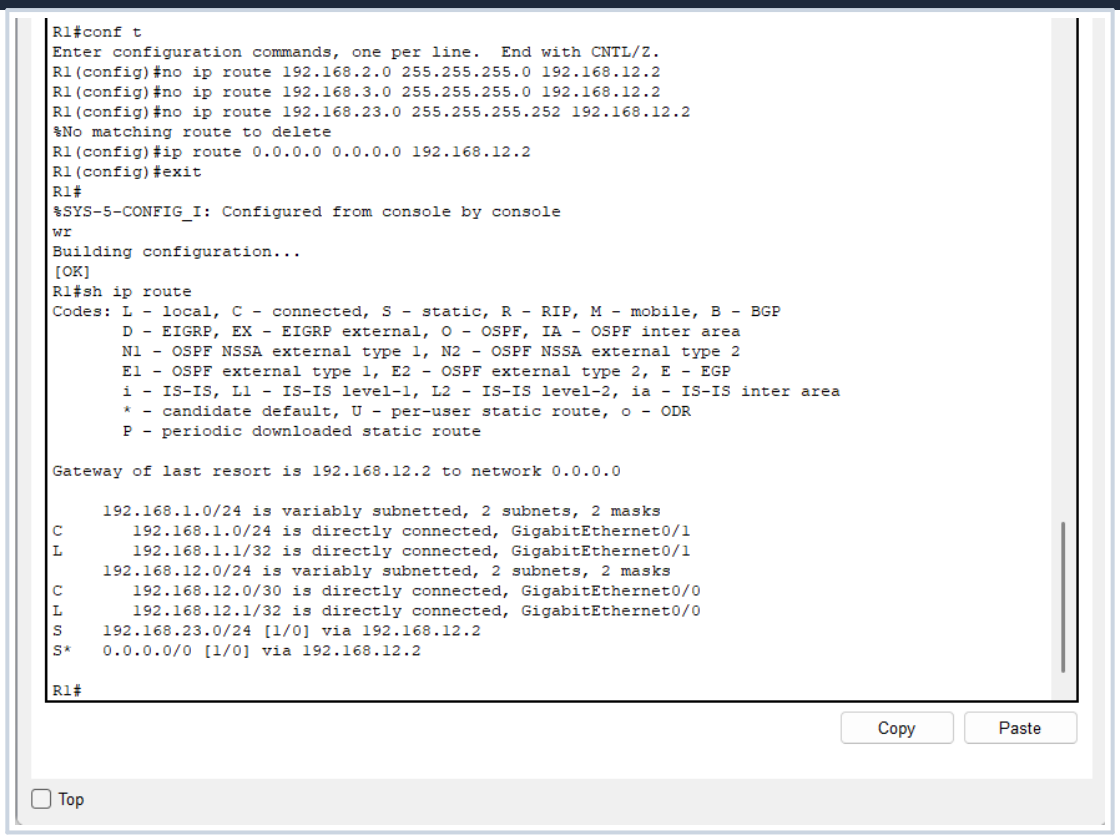
07 Step 4 — Default Route Implementation

Rather than maintaining individual static routes for every remote network, a **default route** acts as a catch-all. It matches any destination not already present in the routing table. This is especially useful for edge/stub routers like R1 that only have one exit path.

On R1, the three individual static routes were removed and replaced with a single default route:

```
R1(config)#no ip route 192.168.2.0 255.255.255.0 192.168.12.2
R1(config)#no ip route 192.168.3.0 255.255.255.0 192.168.12.2
R1(config)#no ip route 192.168.23.0 255.255.255.252 192.168.12.2

R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.12.2
```



```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip route 192.168.2.0 255.255.255.0 192.168.12.2
R1(config)#no ip route 192.168.3.0 255.255.255.0 192.168.12.2
R1(config)#no ip route 192.168.23.0 255.255.255.252 192.168.12.2
%No matching route to delete
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.12.2
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.12.2 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/30 is directly connected, GigabitEthernet0/0
L       192.168.12.1/32 is directly connected, GigabitEthernet0/0
S    192.168.23.0/24 [1/0] via 192.168.12.2
S*   0.0.0.0/0 [1/0] via 192.168.12.2

R1#
```


Figure 14 — R1: Individual statics removed, default route added. Note S 0.0.0.0/0 and 'Gateway of last resort is 192.168.12.2'*

Key Observation: The routing table now shows `S* 0.0.0.0/0 [1/0] via 192.168.12.2`. The asterisk (*) indicates this route is the **gateway of last resort**. R1 will forward ALL unknown traffic to R2, which then makes the more specific forwarding decision.

08 Step 5 — Floating Static Route (Redundancy)

A **floating static route** provides path redundancy by defining a backup route with a higher Administrative Distance (AD). Under normal conditions, the primary route (AD 1) is installed in the routing table. If the primary link fails, the floating route (AD 10) automatically takes over.

A direct link was added between R1 (G0/2) and R3 (G0/2) on the 192.168.13.0/30 network to serve as the backup path:

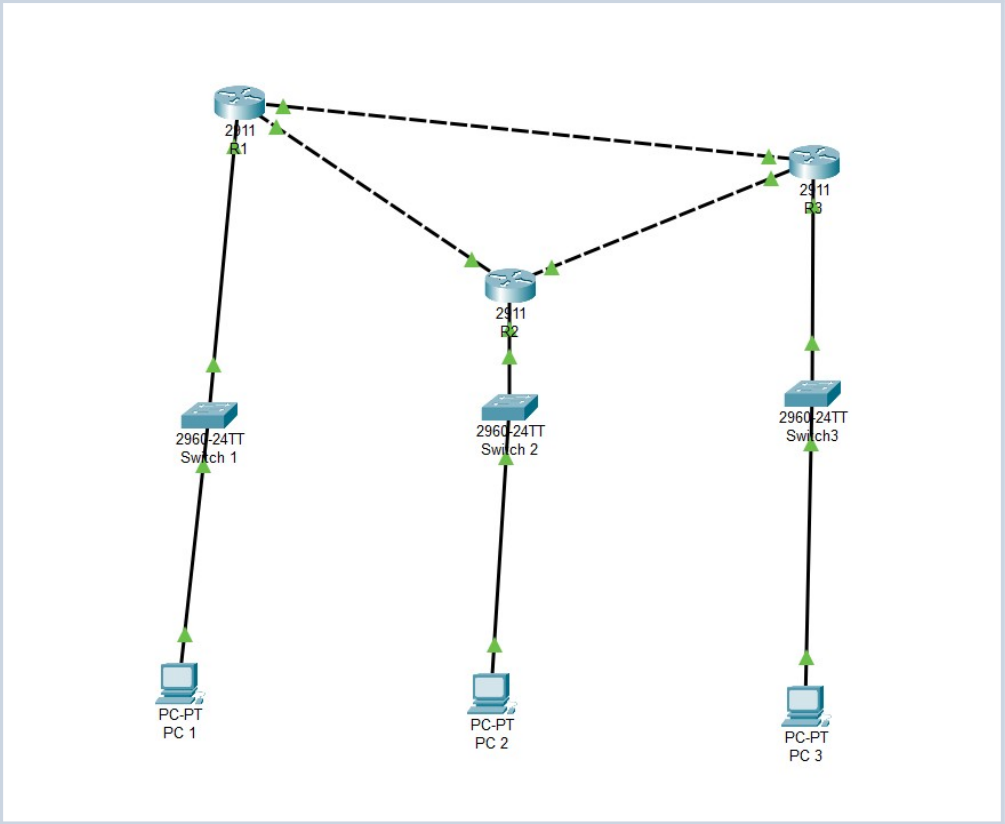


Figure 15 — Updated topology with redundant R1–R3 direct link for floating static route

```
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.12.2      ! Primary (AD=1)
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.13.2 10  ! Floating (AD=10)
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.12.2
R1(config)#ip route 192.168.3.0 255.255.255.0 192.168.13.2 10
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
R1#
```

Figure 16 — R1: Primary route via R2 (AD 1) and floating backup via R3 (AD 10) configured

Route Source	Default AD	Purpose
Directly Connected	0	Interfaces the router owns
Static Route	1	Manually configured — highest preference
EIGRP	90	Cisco proprietary dynamic routing
OSPF	110	Open standard link-state protocol
RIP	120	Legacy distance-vector protocol
Floating Static (this lab)	10	Backup — higher than 1, lower than dynamic

How it works: The floating route (AD 10) remains hidden while the primary (AD 1) is active. If the R1–R2 link goes down, the primary route is removed from the table and the floating route automatically appears — providing seamless failover to the direct R1–R3 path.

09 Results Summary & Key Takeaways

Test / Milestone	Status	Evidence
R1 interface configuration	PASS	G0/0 and G0/1 up/up with correct IPs
R2 interface configuration	PASS	G0/0, G0/1, G0/2 up/up (error corrected)
R3 interface configuration	PASS	G0/0 and G0/1 up/up with correct IPs
Static routes on all routers	PASS	show ip route confirms S entries
PC1 → PC2 ping	PASS	Reply received, 0% loss (after ARP)
PC1 → PC3 ping	PASS	Reply received, 0% loss
PC2 → PC1 and PC3 ping	PASS	Bidirectional reachability confirmed
Traceroute PC1 → PC3	PASS	4 hops: R1 → R2 → R3 → PC3
Default route on R1	PASS	S* 0.0.0.0/0 via 192.168.12.2
Floating static route	PASS	Primary AD 1 + backup AD 10 configured

Key Concepts Demonstrated

Concept	Description
Static Route	A manually configured route: ip route [dest] [mask] [next-hop]. Default AD = 1.
Default Route	The 0.0.0.0/0 catch-all route — gateway of last resort for unknown destinations.
Floating Static	Backup route with elevated AD. Hidden until primary fails, then auto-activates.
Administrative Distance	Trustworthiness metric: lower AD = preferred. Connected=0, Static=1, OSPF=110.
Next-Hop Routing	Specifying the next router's IP (e.g., 192.168.12.2) as the forwarding target.
Bidirectional Routing	Routes must exist in both directions for communication to succeed.

10 Lab Validation Checklist

- ✓ All three routers have correct IP addressing on all interfaces
- ✓ All interfaces show up/up status in show ip interface brief
- ✓ Static routes configured on R1, R2, and R3 for all remote networks
- ✓ PC1 can successfully ping PC2 (192.168.2.x) and PC3 (192.168.3.x)
- ✓ PC2 can successfully ping PC1 and PC3 (bidirectional verification)
- ✓ Traceroute from PC1 to PC3 shows correct hop path: R1 → R2 → R3
- ✓ show ip route on R1 displays S (static) entries with correct next-hops
- ✓ Default route configured on R1: S* 0.0.0.0/0 via 192.168.12.2
- ✓ Gateway of last resort is set on R1
- ✓ Floating static route configured with AD 10 as backup to primary AD 1
- ✓ Understand difference between next-hop IP and exit interface syntax
- ✓ All configurations saved with write memory / copy run start

Lab 4 Complete

Static routing is simple but foundational. OSPF and dynamic routing protocols build on these concepts.